Duality

Lagrangian and dual problem

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Optimization: principles and algorithms

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Lagrangian and dual problem

We provide a rigorous definition of the concepts seen in the previous video.
The objective function including the penalty term is called the Lagrangian
The problem to find the values of the penalty coefficients is the dual problem.

Lagrangian

Primal problem

$$\min f(x) \ \ [f : \mathbb{R}^n \to \mathbb{R}]$$

subject to

$$h(x) = 0 \quad [h : \mathbb{R}^n \to \mathbb{R}^m]$$
$$g(x) \leq 0 \quad [g : \mathbb{R}^n \to \mathbb{R}^p]$$

Lagrangian

$$L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$$
$$\lambda \in \mathbb{R}^m \ \ \mu \in \mathbb{R}^p$$

Dual function

Lagrangian

$$L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$$

Dual function

Lagrangian

$$L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$$

Dual function

$$q : \mathbb{R}^{m+p} \to \mathbb{R} q(\lambda, \mu) = \min_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$$

Dual function

Lagrangian

$$L(x, \lambda, \mu) = f(x) + \lambda^T h(x) + \mu^T g(x)$$

Dual function

$$q : \mathbb{R}^{m+p} \to \mathbb{R} q(\lambda, \mu) = \min_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$$

Note

We want

$$\mu^T g(x) \geq 0 \text{ when } g(x) > 0.$$

Therefore, we impose $\mu \geq 0$.

Dual bound

Theorem 4.5

x^* is an optimal solution of the primal.
If $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^p$, $\mu \geq 0$, then

$$q(\lambda, \mu) \leq f(x^*).$$

Proof

$$\begin{aligned} q(\lambda, \mu) = \min_{x \in \mathbb{R}^n} L(x, \lambda, \mu) \\ \leq L(x^*, \lambda, \mu) \\ = f(x^*) + \lambda^T h(x^*) + \mu^T g(x^*) \\ = f(x^*) + \mu^T g(x^*) \leq f(x^*) \end{aligned}$$

Dual bound

Corollary 4.6

x^* is an optimal solution of the primal.
x is a feasible solution of the primal.
If $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}^p$, $\mu \geq 0$, then

$$q(\lambda, \mu) \leq f(x^*) \leq f(x)$$

Dual problem

Motivation

Find the best lower bound

Dual problem

Motivation

Find the best lower bound

$$\max_{\lambda,\mu} q(\lambda, \mu)$$

Dual problem

Motivation

Find the best lower bound

$$\max_{\lambda,\mu} q(\lambda, \mu)$$

subject to

$$\mu \geq 0$$

Dual problem

Motivation

Find the best lower bound

$$\max_{\lambda,\mu} q(\lambda,\mu)$$

subject to

$$\mu \geq 0$$

and

$$(\lambda,\mu) \in \{\lambda,\mu \mid q(\lambda,\mu) > -\infty\}.$$

Weak duality theorem

If

- x^* is the optimal solution of the primal,
- (λ^*, μ^*) is the optimal solution of the dual, then

$$q(\lambda^*, \mu^*) \leq f(x^*) \,.$$

Duality and feasibility

$$q(\lambda^*, \mu^*) \leq f(x^*)\,.$$

Corollary 4.10

If one problem is unbounded, the other one is infeasible.

Duality and feasibility

		Dual problem		
		Optimal	Unbounded	Infeasible
Primal problem	Optimal		*NO*	
	Unbounded	*NO*	*NO*	*YES*
	Infeasible		*YES*	

Optimality of primal and dual

Corollary 4.11

If $\exists\ x^*$, λ^*, μ^* such that

$$q(\lambda^*, \mu^*) = f(x^*),$$

then they are optimal.

Proof

Consider x primal-feasible and (λ, μ) dual-feasible

$$f(x) \geq q(\lambda^*, \mu^*) = f(x^*) \geq q(\lambda, \mu)$$

Duality and feasibility

		Dual problem		
		Optimal	Unbounded	Infeasible
Primal problem	Optimal	*YES*	NO	*NO*
	Unbounded	NO	NO	YES
	Infeasible	*NO*	YES	*??*

Summary

The primal and the dual problems are strongly related.
The dual function is a lower bound on the optimal value of the primal problem.
The dual problem consists in finding the best lower bound.